Final Annotation Codebook

1 Overview

This document reports the final annotation codebook used in the paper, together with the reduced label set adopted for dataset construction and baseline modeling.

The annotation protocol was designed to prioritize directly observable events and states, avoiding categories based on strong psychological interpretations, diagnoses, or unverifiable inferences. For clarity, the original ELAN annotation structure is reported here through a logical tier grouping used consistently in the paper.

2 Logical Tier Mapping

Logical tier group	Original ELAN tier(s)
Individual tiers	INDIVIDUAL, INDIVIDUAL_2, INDIVIDUAL_3
Relational tiers	RELATIONAL, RELATIONAL_2
Support context tier	SUPPORT_CONTEXT
Tag note tier	TAG_NOTE

Table 1: Mapping between the logical tier groups used in the paper and the original ELAN tiers.

3 Full Annotation Codebook

3.1 Individual tiers

A1 — Stands up / sits down

Definition. Clear transition between sitting and standing position, or vice versa.
When to use. When the person clearly leaves the seated posture or clearly returns to it.
When not to use. Do not use for small postural adjustments on the chair.
Unit. Person.
Temporal type. Event.

A2 — Enters / exits

Definition. Visible entrance into or exit from the room or from the central interaction area.
When to use. When a person clearly enters or leaves the relevant interaction space.
When not to use. Do not use for small internal displacements within the room.
Unit. Person.
Temporal type. Event.

A3 — Closed posture

Definition. Maintained closed, withdrawn, or strongly folded posture, for example leaning over the table.
When to use. When the posture is maintained for at least 5 consecutive seconds.

When not to use. Do not use for a brief or purely functional forward bend.
Unit. Person.
Temporal type. State.

A4 — Disengagement from the group

Definition. Head and/or body stably oriented away from the group or the ongoing activity.
When to use. When orientation away from the group is maintained for at least 5 consecutive seconds.
When not to use. Do not use for a brief glance away.
Unit. Person.
Temporal type. State.

A5 — Repetitive movement

Definition. Recurrent body movement with a repeated and self-similar pattern.
When to use. When a similar motor pattern repeats at least 3 times in 10 seconds, or remains visible for at least 5 consecutive seconds.
When not to use. Do not use for a single gesture or a simple functional movement.
Unit. Person.
Temporal type. State or episode.
Note. Used descriptively and cautiously, without clinical inference.

3.2 Relational tiers

B1 — Proximity

Definition. Two people remain in close spatial proximity for a non-trivial interval.
When to use. When proximity is maintained for at least 5 seconds and does not appear purely accidental.
When not to use. Do not use when seating arrangement makes proximity inevitable and no further relational evidence is visible.
Unit. Dyad.
Temporal type. State.

B2 — Affiliative contact

Definition. Positive bodily contact, such as affectionate, supportive, or clearly playful touch.
When to use. When contact is visibly affiliative or reassuring.
When not to use. Do not use for accidental or purely functional contact.
Unit. Dyad.
Temporal type. Event or episode.

B3 — Ambiguous contact

Definition. Visible bodily contact whose relational meaning cannot be decided confidently.
When to use. When contact is visible but its meaning remains uncertain.
When not to use. Do not use when the meaning is clearly affiliative or clearly non-relational.
Unit. Dyad.
Temporal type. Event.

B4 — Mutual orientation

Definition. Two people orient head and/or body toward each other simultaneously or almost simultaneously.
When to use. When reciprocal orientation is maintained for at least 2 consecutive seconds.
When not to use. Do not use when the alignment is too brief or too uncertain.
Unit. Dyad.

Temporal type. Short state or episode.
Note. Used as a cautious proxy for interpersonal visual/bodily orientation.

B5 — Possible tension

Definition. Interaction suggesting possible tension, protest, friction, or problematic exchange.
When to use. When visible behavioral cues support a cautious reading of tension.
When not to use. Do not use when the interpretation is only impressionistic and not grounded in visible behavior.
Unit. Dyad.
Temporal type. Episode.

B6 — Attention-triggered reaction

Definition. Visible behavioral change following mutual orientation or an attention cue toward another person.
When to use. When an orientation cue is followed by a readable behavioral reaction.
When not to use. Do not use when the sequence between cue and reaction is not clearly readable.
Unit. Dyad.
Temporal type. Episode.

3.3 Support context tier

C1 — Educational support

Definition. Focused educator intervention directed at supporting or accompanying a person.
When to use. When an educator is visibly engaged in a targeted supportive action.
When not to use. Do not use for generic interaction with the whole group.
Unit. Dyad.
Temporal type. Episode or short state.

C2 — Co-regulation

Definition. Educator intervention aimed at calming, regulating, or helping a person re-enter the activity.
When to use. When the intervention is clearly regulatory rather than generic.
When not to use. Do not use when the interaction is too brief or indistinguishable from ordinary communication.
Unit. Dyad.
Temporal type. Episode.

3.4 Tag note tier

D1 — Poor visibility

Definition. The scene does not allow a sufficient reading of the observed behavior.
When to use. When the body is occluded, out of frame, or otherwise poorly visible.
Unit. Person or scene.
Temporal type. Tag.

D2 — Ambiguous meaning

Definition. The behavior is visible, but its meaning cannot be assigned confidently.
When to use. When the scene should be preserved as ambiguous rather than over-interpreted.
Unit. Any level.
Temporal type. Tag.

D3 — Possible discomfort

Definition. The scene suggests possible pain, fatigue, or discomfort without confirmation.
When to use. When visible cues suggest discomfort but do not support stronger claims.
When not to use. Do not use as a clinical or diagnostic statement.
Unit. Person.
Temporal type. Tag or episode.

4 Reduced Label Set Used for Dataset Construction

For dataset construction and baseline modeling, the full annotation set was reduced to a smaller set of concise labels.

Reduced label	**Included annotation label(s)**
Stand	A1_stands_up
Sit	A1_sits_down
ClosedPosture	A3_closed_posture
DisengagedRepetitive	A4_disengagement_from_group + A5_repetitive_movement
SocialOrientation	B1_proximity + B4_mutual_orientation
AffiliativeContact	B2_affiliative_contact

Table 2: Reduced label set derived from the full annotation codebook.

The baseline person-level classification experiments reported in the paper were conducted on the four individual labels Stand, Sit, ClosedPosture, and DisengagedRepetitive. The two relational aggregates SocialOrientation and AffiliativeContact were retained for completeness in the dataset description.

5 Interpretation Note

The annotation codebook was designed for descriptive observation and computational modeling. Labels should therefore be interpreted as operational descriptions of visible behaviors and interaction patterns, rather than as diagnostic or clinical categories.